A High Performance Muscle-like Actuator using Multi-layer Electrostatic Film Motors

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***Abstract*—Artificial muscles have been researched for decades; however, most of them have difficulty in control with high accuracy due to non-linearity, such as hysteresis and creep phenomenon. On the other hand, electrostatic film motors have fine controllability, as well as high performance, flexibility and high accuracy, and are excepted to be used in robotics. The rigid prototype developed in a previous work had low force-to-weight ratio due to its bulky structure, while another flexible prototype only provided 4.5 N due to its small size and few layers. This paper reported a new muscle-like actuator with high force-to-weight ratio using multi-layer electrostatic film motors, in which 11 pairs of double-sided drive films were assembled together. Experiments were carried out under two conditions, static and dynamic, in which the force performance was evaluated using a load cell, or by lifting a weight, respectively. The prototype weighing 235 g recorded a force of 163 N under voltage of 1.8 kV and lifted around 4 kg weight under voltage of 1.3 kV. Compared with the existing structure with similar sizes, the prototype in this work has 4.7 times larger force-to-weight ratio.**

***Index Terms*—electrostatic film motor, synchronous-type motor, muscle-like actuator**

I. Introduction

Soft actuators, especially artificial muscles, have evolved over decades, and are expected to be used in robotics to realize safe human-machine interaction in physical perspectives. Since robots have begun to intervene in social common life with new roles, such as surgery robot [1], restaurant service robot [2], healthcare robot [3], and education robot [4], safe interaction with robots has attracted increasing attention. Artificial muscles with excellent flexibility and high performance, have been regarded as the most potential candidates to increase interaction safety of robots [5], [6]. Artificial muscles are materials or devices that are similar with natural muscle, and could achieve desired motions by combining three basic motions including contraction, expansion and rotation within a single component due to an external stimulus, such as temperature, voltage, currents, and pressure [7]–[9]. Compared with electromagnetic machines that are commonly used in conventional robots, artificial muscles have some unique characteristics, such as simple structure, light weight and mechanical flexibility with high performance. However, the main disadvantage of artificial muscles is the hysteresis and creep phenomenon during operation, which makes it difficult to realize position/force control with high accuracy [10]. Although artificial muscles are widely applied in humanoid robots and soft robots, they are not good options to replace the electromagnetic machines from those robots with requirement of high accuracy.

On the other hand, electrostatic film motors with characteristics of flexibility, lightness in weight and high performance, could realize high-precision motion control and are also expected to be used as "muscles-like" actuators in robots [11]–[13]. Similar to electrostatic motors in micro-electromechanical systems (MEMS), electrostatic film motors are also driven by electrostatic force between charges on the surfaces of slider/rotor and stator. Differently from MEMS motors, fine-patterned electrodes of electrostatic film motors are printed in plastic films or thin boards by utilizing flexible printed circuit (FPC) technology [14]. As a result, electrostatic film motors have the characteristics of mechanical flexibility and lightness in weight, which is similar to artificial muscles. Generally, electrostatic film motors are operated in dielectric liquid to avoid discharge in air, and can provide high performance by applying high voltage. Since electrostatic film motors do not have the problems of hysteresis and creep, the position/force could be controlled with higher accuracy than typical artificial muscles. Furthermore, as the impact energy generated by accidental collision could be absorbed by the flexible films and the dielectric liquid, it is possible to achieve safe interaction between robots and humans. Considering that electrostatic film motors have the advantages of both flexibility of artificial muscles and the high controllability of electromagnetic motors, they seem a promising candidate for driving robots.

In previous research, a muscle-like actuator was fabricated

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by using single-sided drive electrostatic film motors, in which only one surface of the films could be used for driving [11]. Although the prototype provided large thrust force up to 310 N (at around 1.4 kV), mechanical flexibility was not realized due to its rigid and bulky structure. To double the performance of multi-layer actuators, double-sided drive film, in which the distances between electrodes and top/bottom surfaces of a film are almost the same, was proposed [13]. On the other hand, the electrostatic film motors can be packaged in a plastic bag instead of a rigid box [15]. Although the flexible prototype in [15] could realize the mechanical flexibility and interaction safety, it can only generate 4.5 N at 2 kV_{0-p}. Therefore, this work tries to fabricate a high performance muscle-like actuator with mechanical flexibility by integrating the existing techniques from the previous studies. In the prototype, double-sided drive electrostatic film motors with similar structure in [13] were stacked and were encapsulated in a plastic bag as in [15].

The rest of this paper is structured as follows: Section II briefly introduces the principle of the synchronous electrostatic film motor. Section III introduces the structure of the muscle-like actuator in detail. The performance of the muscle-like actuator is estimated in experiments, and the results are described and discussed in section IV. The final section is conclusion.

II. Electrostatic Film Motor

A. Principle of Electrostatic Film Motor

Electrostatic film motors are made up of slider films and stator films. In typical slider/stator films, 3-phase parallel electrodes are printed between two layers of polymer films by using FPC technology. There is a capacitance relationship between any two phases in the motor, which could be expressed as a capacitance matrix [16]. When a 3-phase sinusoidal high-voltage is applied to the motor, it forms voltage potential wave on each film. As a result, electrostatic attraction force between the voltage potential waves is generated to propel the slider film to move synchronously. Using the capacitance matrix described in [17], [18], the excited thrust force is expressed as

$$F = kC_{m1}\sin(\theta_x - 2\phi + \frac{\pi}{3}) + 2kC_{m2}\sin(2\theta_x + 2\phi - \frac{\pi}{3}) \quad (1)$$

where the coefficient $k = 3\pi V_0^2/2p$ depends on the voltage amplitude V_0 (of phase voltage) and the electrode pitch p of the motor, $C_{m1/m2} = AC_{a1/a2}$ depend on capacitance constants per unit area $C_{a1/a2}$ of the motor and effective actuation area A, which could be multiplied by stacking films and be doubled by using double-sided drive films, ϕ is the phase of the applied voltage, $\theta_x = 2\pi x/3p$ denotes the electric angle of the slider film position x.

B. Performance

Thrust force is a key parameter in evaluating the motor performance. The relation between the thrust force F and the phase ϕ under different slider positions is depicted in the upper plot of Fig. 1, in which thrust forces were normalized. The black asterisks represent the peak forces F_{peak} at each slider position. It is observed that peak force F_{peak} changes with the slider position θ_x, which was plotted in the lower figure. The maximum and minimum values of F_{peak} are denoted as F_{pmax} and F_{pmin}, respectively. Therefore, to fully evaluate the motor performance, peak force is required to be estimated under different positions in experiments.

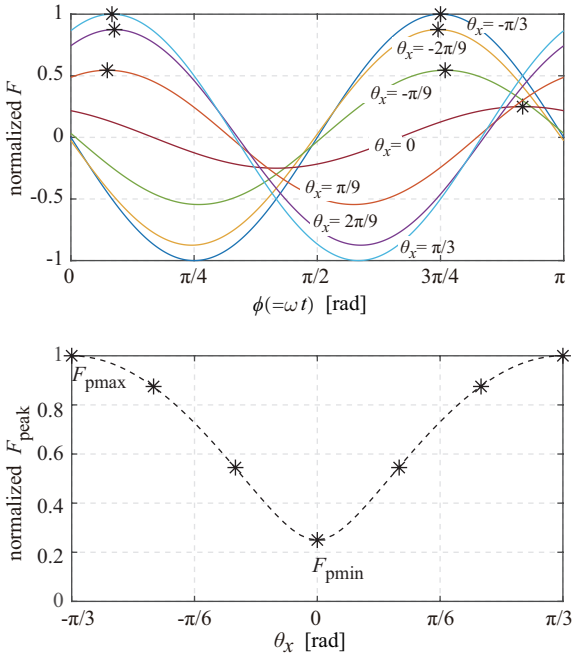


Fig. 1. Relation between thrust force F and phase ϕ under different slider position θ_x, and the peak force ripple within an electrode pitch

III. Muscle-like Actuator

A. Configurations of Effective Actuation Area

Effective actuation area, which is the overlapping area between electrode areas of the slider/stator films, affects the motor performance. According to the configuration of effective actuation area, electrostatic film motors could be divided into "sliding type" and "contracting type" [15]. In the sliding type, a smaller slider film reciprocates on a larger stator film with keeping a constant effective actuation area. As a result, force characteristics remain constant regardless of the slider position.

In the contracting type, on the other hand, the effective actuation area changes in a linear relationship with the slider position x. Therefore, peak forces F_{pmax}/F_{pmin} also change with the slider position. As the characteristics of the contracting type are closer to that of biological/artificial muscle, it was adopted in this work.

B. Structure

A high performance muscle-like actuator was fabricated using the contracting type electrostatic film motors, as shown in Fig. 2, in which silder/stator films share the same structure.

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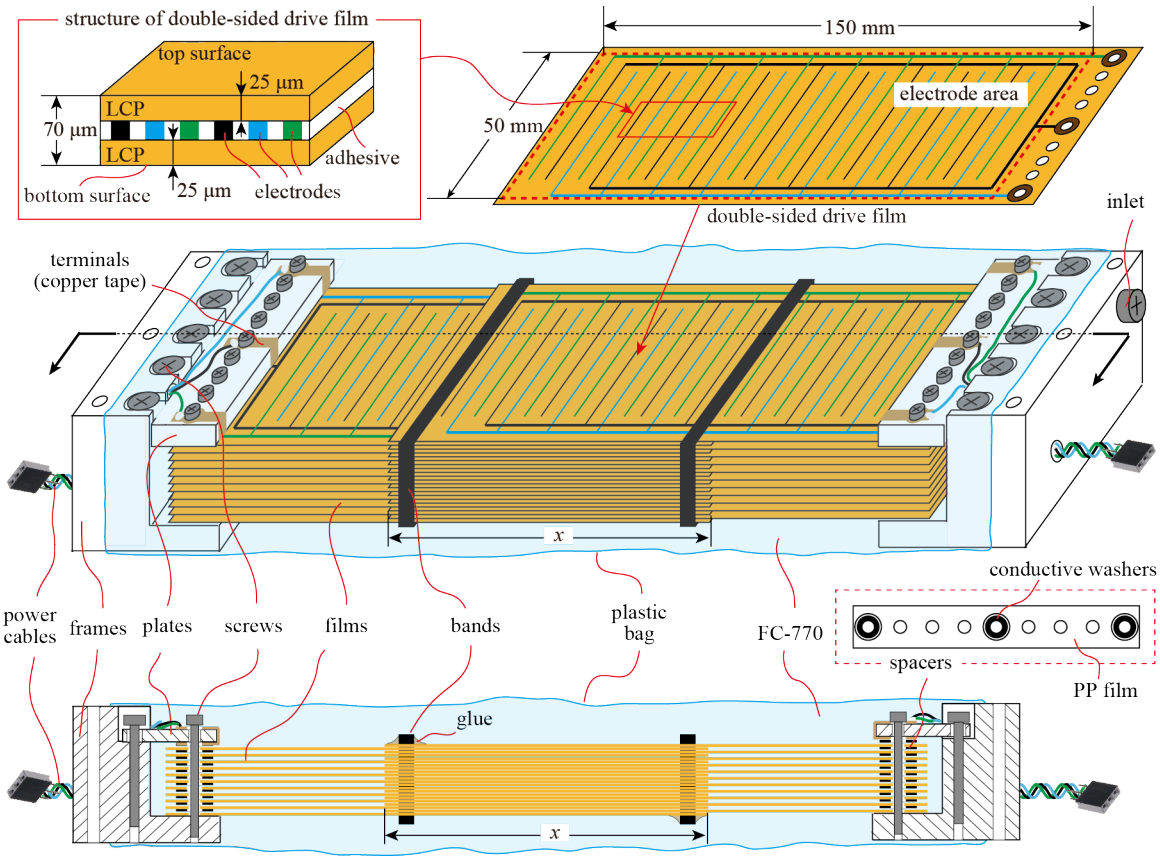
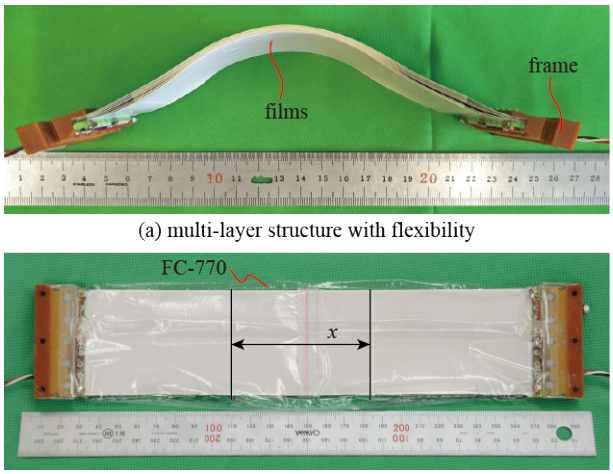


Fig. 2. Schematic diagram of the muscle-like actuator module with multi-layer structure



(a) multi-layer structure with flexibility

(b) sealed prototype

Fig. 3. Prototype of the muscle-like actuator module, in which 11 pairs of electrostatic film motor are stacked together.

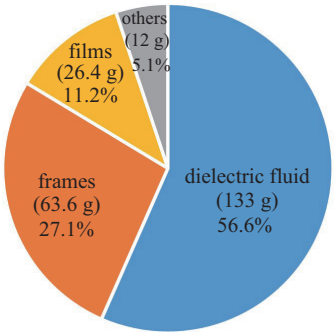


Fig. 4. Percentage contribution of components to muscle-like actuator with a mass of 235 g

The film has an electrode area of 50 mm×150 mm with an electrode pitch of 200 μm. The distance from the electrode to the film surface was designed to be 25 μm for the top and bottom surfaces, respectively, which allows the film to generate thrust force on its both surfaces. In slider/stator module, there were 11 layers of films stacked and fixed between the frames and plates by using screws, while spacers with conductive washers were placed between films to maintain spacing gap and conduction. Terminals that were made of copper tape were attached to the plate surface, for connecting the 3-phase electrodes and power cables. These plates were fixed with frames by using larger screws. The power cables passed through an oblique hole, and then were soldered on the terminal surface. The oblique hole was sealed with glue after finishing the whole structure. The slider and stator modules were assembled layer-by-layer with spreading glass beads between adjacent films. Bands that were folded into rectangular rings, were

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fixed on the top/bottom surfaces to prevent slider/stator films from rotating. Finally, the multi-layer muscle-like actuator was sealed in a plastic bag (polyethylene, thickness 40 μm) and dielectric fluid (3M, Fluorinert FC-770) was injected through an inlet hole.

The prototype of the muscle-like actuator before/after packaging with a plastic bag is shown in Fig. 3. The initial size of the muscle-like actuator is 280 mm $\times$ 60 mm $\times$ 12 mm. During operation, the length of the actuator is possible to be shrunk to 200 mm with a stroke of 80 mm. Considering the double-sided drive films and the multi-layer structure, the total effective actuation area of the muscle-like actuator could be expressed as $(2n-1)\times x$ mm$\times$50 mm, where n is the number of film pairs which is 11 in this work, x is the length of the overlapping electrode area whose value is in the range of [70, 150] mm. The percentage contribution of components to this muscle-like actuator prototype is shown in Fig. 3. Dielectric fluid FC-770 occupies 56.6% of the prototype weight, which is about 133 g. However, the weight of all films is 26.4 g, which is only 11.2% of the total weight.

IV. Performance Estimation in Experiments

A. Experiment Setup

To evaluate the performance of the muscle-like actuator that was introduced in the previous section, series of experiments have been carried out. The experiment setup is shown in Fig. 5. Three-phase sinusoidal-wave high voltage, which was generated by a digital signal processing system (Dspace, ds1104) and was amplified by a set of high-voltage amplifiers (NF, HVA4321), was applied to the muscle-like actuator. The muscle-like actuator was evaluated under static and dynamic conditions.

In the static condition, the thrust force was measured at various positions under different voltages to obtain the peak force distribution of the muscle-like actuator. The stator was fixed with a base plate, while the slider was connected with a load cell (Kyowa, LUR-A-1kNSA1) which was fixed on a position stage and was used to measure the thrust force, as shown in Fig. 5(a). The thrust force was measured at the position where x was around 140 mm. As peak force ripples with the slider position, the thrust force was measured under different positions with an interval of 10 μm in a range of an electrode pitch 200 μm. Since an electrode pitch 200 μm is much smaller than the length $x = 140$ mm, the overlapping electrode area was regarded as a constant in the experiments. The amplitude of the applied voltage changed from 0.5 kV to 1.8 kV. The frequency was 2 Hz. As a result, the peak force distribution under different voltages and various slider positions could be obtained.

In the dynamic condition, the muscle-like actuator was lifted a weight at a constant velocity (2.4 mm/s) to evaluate the load capability. The stator was fixed on a base plate, while the slider was connected with a stainless steel wire, that was used to lift the weight through a pulley, as shown in Fig. 5(b). Considering that enough space is needed when lifting the weight, the position where x takes the minimum value of

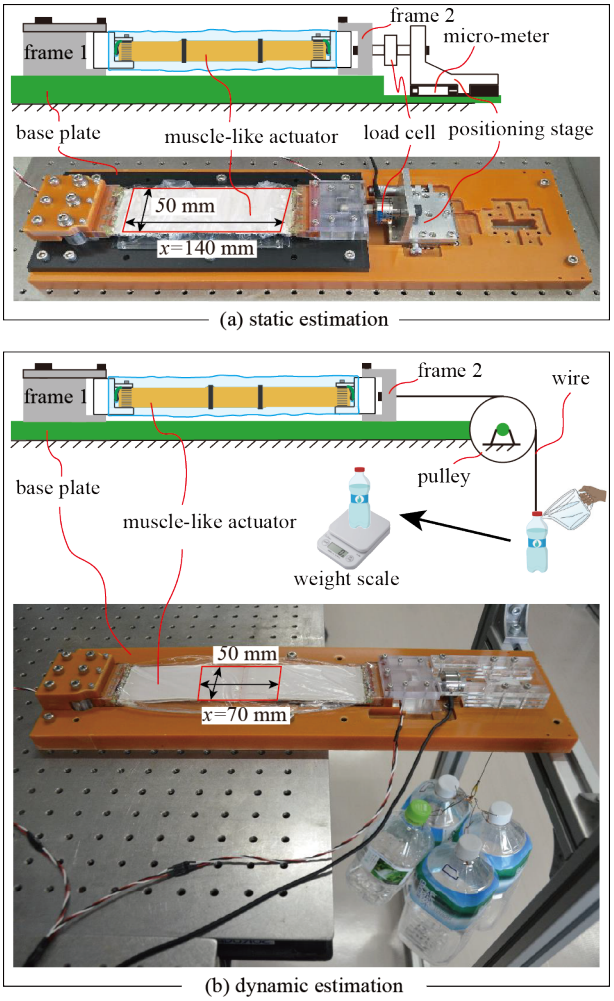


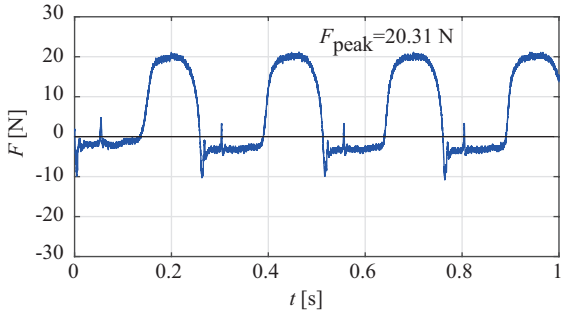
Fig. 5. Experiments setup

70 mm was set as the initial lifting position. Water bottles were used as the weight, due to its convenience of adjusting the amount of water. The maximum weight of the water bottles that the actuator could successfully lift is used to describe the load capability, which was measured under different voltages. The amplitude of the applied voltage changed from 0.5 kV to 1.3 kV. The frequency was 2 Hz.

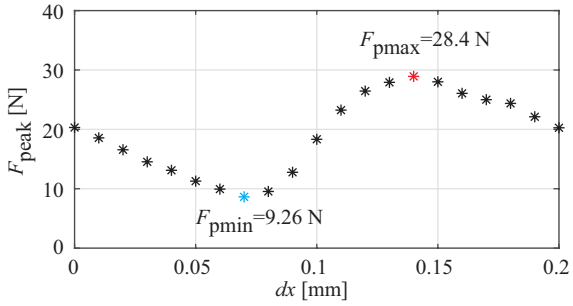
B. Results and Discussion

The experiment results under the static condition are shown in Fig. 6. The slider displacement is expressed by using dx. Figure 6(a) is the thrust force that was measured at the initial slider position ($dx = 0$ mm) under 0.6 kV_{0-p}. It is observed that only contraction force (positive direction) was accurately measured while the stretch force (negative direction) can not be obtained due to the buckling of the films. Peak force F_{peak} under different slider positions is shown in Fig. 6(b) using asterisks, in which the max/min peak force are marked using red/blue asterisks. It is confirmed that force ripple exists in the muscle-like actuator as same as typical electrostatic film motors, with the peak force changing in the range of

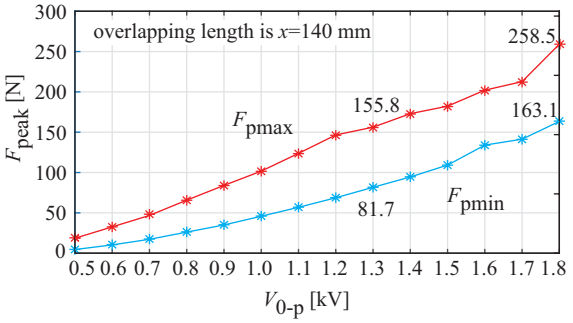
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(a) measured thrust force at initial position (dx=0 mm)



(b) peak force ripple over the slider positio (at 0.6 $kV_{0\text{-}p}$)



(c) F_{peak} under different voltages

Fig. 6. Measured thrust force F, peak force F_{peak} and its relation with the applied voltage

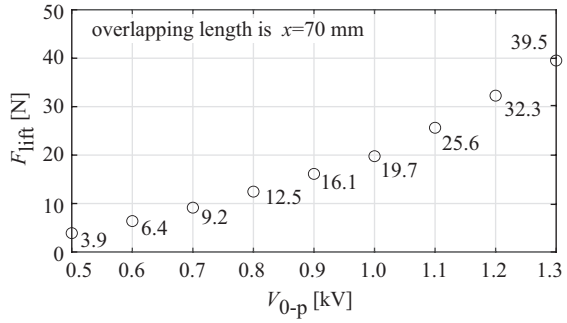


Fig. 7. Maximum lifting load under various voltages

[F_{pmin}, F_{pmax}]. The relation between the max/min peak force and the applied voltage is shown in Fig. 6(c) using red/blue lines. The peak forces under various slider positions and different voltages are distributed between the red line and the blue line.

The experiment results in the dynamic condition are shown

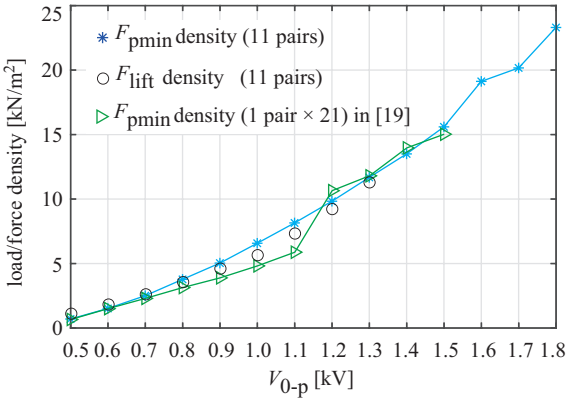


Fig. 8. Relationship between the load/force density and applied voltage under different conditions

in Fig. 7. It is observed that a weight of about 4 kg was lifted by the muscle-like actuator when the voltage amplitude was 1.3 kV. In the static condition, however, the peak force under 1.3 kV is in the range of [81.7, 155.8] N, which is much larger than $F_{lift} = 39.5$ N in the dynamic condition. That is because the actuator were operated under different overlapping areas in two conditions.

To compare the load capability in Fig. 7 with the detected thrust force in Fig. 6(c), load/force density under the same unit of kN/m^2 were calculated by dividing load/force with corresponding overlapping areas, which are plotted together in Fig. 8. Blue star marks and blue line express the force density of F_{pmin} in Fig. 6(c), while circle marks are the load density of F_{lift} in Fig. 7. It is observed that the maximum lifting load F_{lift} does not depend on the maximum peak force F_{pmax} but coincide with the minimum peak force F_{pmin}. The force density under various voltages of a basic single-layer motor in [19] was also transformed by being divided with overlapping area and was plotted in Fig. 8 using green triangle marks and green line. It is confirmed that the output of the multi-layer actuator is $2n-1$ (n is 11 pairs) times that of a single layer, due to the double-sided drive films used in this work.

Compared with the previous research, the prototype reported in this paper has excellent performance, as shown in Table I. In "structure 1", the multi-layer structure was encased in a rigid box and the total mass was up to 3.6 kg [11]. Although "structure 2" was encapsulated in a plastic bag, its force-to-weight ratio was in a low level, as it only provided several newtons due to its single-sided drive films with fewer layers [15]. The force-to-weight ratio of the prototype in this paper is 4.7 times of that of "structure 1" and 11.2/11.7 times of that of "structure 2" due to its light package and double-sided drive films.

V. Conclusion

This paper reported a muscle-like actuator with high performance, in which 11-pairs of double-sided drive electrostatic film motors with thin structure were assembled and were encapsulated in a plastic bag that is filled with dielectric

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TABLE I
KEY METRICS OF THE MULTI-LAYER ELECTROSTATIC FILM MOTORS IN THIS WORK AND IN LITERATURE

	prototype in this work	structure 1 in [11]	structure 2 in [15]
F_{pmin} (at V_{o-p}=1.4 kV) [N]	94.5	310	1.8 (flexible)/3.3 (rigid)
size [mm]	$280 \times 55 \times 12$	$210 \times 170 \times 80$	$130 \times 35 \times 5$
weight [g]	235	3600	50 (flexible)/96 (rigid)
force-to-weight ratio (at V_{o-p}=1.4 kV) [N/kg]	402.1	86.1	36 (flexible)/34.4 (rigid)
electrode pitch [μm]	200	200	200
No. of layers	11	50	3
driving type	double-sided drive	single-sided drive	single-sided drive
Flexibility/Rigidity	flexible	rigid	flexible/rigid

fluid. The force ripple that exists in the theory of the regular synchronous-type electrostatic film motors was visualized in Fig. 1, and was confirmed by the experiment results. The force-voltage characteristics of the actuator were measured under two conditions, whose results confirmed that the capability of electrostatic film motors depends on the minimum value rather than the maximum value of the peak forces under various slider positions. The experiment results also confirmed that the reported prototype could provide $2n-1$ (n is the number of film pairs) times force of the basic single-layer electrostatic film motor. Compared to the previous structure with the similar size, the prototype in this work has 4.7 times force-to-weight under the same voltage.

It should be noted that there are still many challenges in the practical application of this prototype. The proportion of the films in the overweight is only about one-fourth, which is possible to be increased by adding more layers of the films. However, the number of films in the multi-layer structure is limited by the assembly precision and the strength of the frames. The disturbance generated by the dielectric liquid will affect the dynamic performance of the actuator, which puts a high demand on the control algorithm. In addition, the dielectric fluid evaporated over time and caused expansion of the plastic bag, which considerably affects the performance of the actuator. These challenges should be investigated in the future. Considering the relation between F_{pmin} and F_{lift}, it implies that motors with skewed-electrode [16] potentially have higher load capability than the actuators with regular-electrode like in this paper, which should also be studied in future research.

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